

Chapter 6

Capacitors and Inductors



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Lec#1

6.2 Capacitors

6.3 Series and Parallel Capacitors

Lec#2

6.4 Inductors

6.5 Series and Parallel Inductors



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Differentiation $\frac{d}{dt}$: is a tool that we shall use to find gradients of graphs.

Let $U=U(t)$, $V=V(t)$,
and a – constant

1. $\frac{d}{dt}(a) = 0$

Consider $v(t) = 5 V$,

Find $\frac{d}{dt}v(t) = v'(t)$

$$\begin{aligned}\frac{d}{dt}v(t) &= \frac{d}{dt}5 \\ &= 0\end{aligned}$$

2. $\frac{d}{dt}(aU) = a \frac{dU}{dt}$

Consider $i(t) = 5t - 25 \text{ mA}$, Find $\frac{d}{dt}i(t)$

$$\begin{aligned}\frac{d}{dt}i(t) &= \frac{d}{dt}(5t - 25) \\ &= 5 \frac{d}{dt}t - \frac{d}{dt}25 \\ i'(t) &= 5 \text{ m}\end{aligned}$$

3. $\frac{d}{dt}(aU^n) = n aU^{n-1}$

Consider $i(t) = 5t^3 - 2.5t \text{ A}$,

Find $\frac{d}{dt}i(t)$

$$\begin{aligned}\frac{d}{dt}i(t) &= \frac{d}{dt}(5t^3 - 2.5t) \\ &= 5 \frac{d}{dt}(t^3) - 2.5 \frac{d}{dt}t \\ i'(t) &= 15t^2 - 2.5\end{aligned}$$



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Differentiation

4. $\frac{d}{dt}(\sin U) = \cos U \frac{dU}{dt}$

Consider $v(t) = 5 \sin 0.25t \text{ V}$,

Find $\frac{d}{dt}v(t)$

$$\begin{aligned}\frac{d}{dt}v(t) &= \frac{d}{dt}(5 \sin 0.25t) \\ &= 5 \cos 0.25t \frac{d}{dt}(0.25t) \\ &= 5 \cos 0.25t (0.25) \\ v'(t) &= 1.25 \cos 0.25t\end{aligned}$$

5. $\frac{d}{dt}(\cos U) = -\sin U \frac{dU}{dt}$

Consider $f(t) = -2.5 \cos (12t^2 - 5)$,

Find $\frac{d}{dt}f(t)$

$$\begin{aligned}\frac{d}{dt}f(t) &= -2.5 \frac{d}{dt}\cos (12t^2 - 5) \\ &= -2.5 \{-\sin (12t^2 - 5)\} \frac{d}{dt}(12t^2 - 5) \\ &= 2.5 \sin (12t^2 - 5) (24t) \\ f'(t) &= 60 t \sin (12t^2 - 5)\end{aligned}$$

6. $\frac{d}{dt}e^U = e^U \frac{dU}{dt}$

Consider $V(t) = 5e^{-3t} - 2.5t^2 \text{ V}$,

Find $\frac{d}{dt}V(t)$

$$\begin{aligned}\frac{d}{dt}V(t) &= \frac{d}{dt}\{5e^{-3t} - 2.5t^2\} \\ &= 5 \frac{d}{dt}(e^{-3t}) - 2.5 \frac{d}{dt}t^2 \\ V'(t) &= -15e^{-3t} - 5t\end{aligned}$$



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Integration: The integral sign $\int$ represents **integration**.

It describe displacement, area, volume, and other concepts under the curve

Let $U=U(t)$, $V=V(t)$, and a, c – constant

1. $\int a \, dt = at + c$

Consider $v(t) = 5 \, V$, Find $\int v(t) \, dt$

$$= \int 5 \, dt$$

$$= 5t + c$$

2. $\int a t^n \, dt = a \frac{t^{n+1}}{n+1} + c \quad n \neq -1$

Consider $i(t) = 5t^3 - 25 \, mA$, Find $\int i(t) \, dt$

$$= \int (5t^3 - 25) \, dt$$

$$= \int 5t^3 \, dt - \int 25 \, dt$$

$$= 5 \frac{t^4}{4} - 25t + c$$

3. $\int \sin at \, dt = -\frac{1}{a} \cos at + c$

Consider $v(t) = 5 \sin 0.25t \, V$,

Find $\int v(t) \, dt$

$$= 5 \int \sin 0.25t \, dt$$

$$= -\frac{5}{0.25} \cos 0.25t + c$$

4. $\int \cos at \, dt = \frac{1}{a} \sin at + c$

Consider $f(t) = 60 \sin (12t-5)$,

Find $\int f(t) \, dt$

$$= 60 \int \sin(12t-5) \, dt$$

$$= \frac{60}{12} \cos(12t-5) + c$$



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Integration:

5. $\int e^{at} \, dt = \frac{1}{a} e^{at} + c$

Consider $V(t) = 5e^{-3t} - 2.5t^2 \, V$,

Find $\int V(t) \, dt$

$$= \int (5e^{-3t} - 2.5t^2) \, dt$$

$$= 5 \int e^{-3t} \, dt - 2.5 \int t^2 \, dt$$

$$= \frac{5}{-3} e^{-3t} - 2.5 \frac{t^3}{3} + c$$

Consider a function $v(t)$

Derivative of $v(t)$ is $\frac{d}{dt} v(t)$

$$= v'(t)$$

Also Integral of $v'(t)$

$$= \int v'(t) \, dt = v(t)$$



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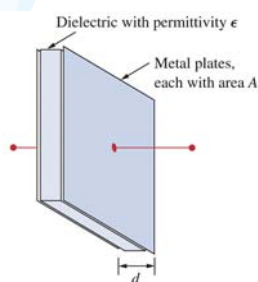
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6.2 Capacitors

Physical characteristics

- A capacitor is a passive element designed to store energy in its **electric field**.
- A capacitor consists of two conducting plates separated by an insulator (or dielectric).



$$C = \frac{\epsilon A}{d}$$

A = Surface area of the metal plate [m^2]

ϵ = permittivity of the dielectric [F/m]

d = distance between plates [m]

$\epsilon = \epsilon_r \epsilon_0$

ϵ_r = Relative permittivity of material

ϵ_0 = Permittivity of air ($8.854 \times 10^{-12} \text{ F/m}$)



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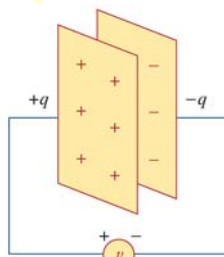
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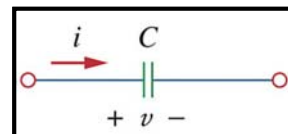
6.2 Capacitors

Electrical characteristics

- Capacitance C is the ratio of the charge q on one plate of a capacitor to the voltage difference v between the two plates, measured in farads (F).



$$C = \frac{q}{v}$$



$$q(t) = Cv(t)$$

$$\frac{d}{dt}q(t) = \frac{d}{dt}(Cv) = C \frac{d}{dt}v(t)$$

$$i(t) = C \frac{d}{dt}v(t)$$

If $i > 0$: Capacitor is charging.
If $i < 0$: Capacitor is discharging.

OR

$$v(t) = \frac{1}{C} \int_{-\infty}^t i(t) dt = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$



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6.2 Capacitors

Energy and Power

$i = C \frac{dv}{dt}$

- The energy, **w**, stored is: $w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$
- power, **p**, in the capacitor is: $p = vi = vC \frac{dv}{dt}$
- Important characteristics of capacitors:
 - It is an **open circuit** to dc ($dv/dt = 0$).
 - Its voltage **cannot change abruptly** ($dv/dt = \infty$).
 - Does not dissipate energy, but stores it when charging or provides it when discharging.



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6.2 Capacitors

New Equations

V-I characteristic of a Capacitor

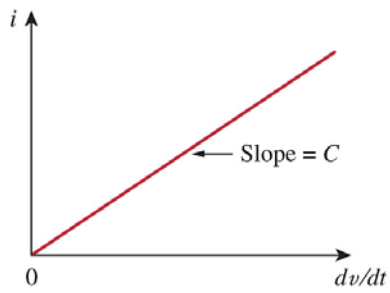
$$i = C \frac{dv}{dt}$$

$$q = Cv$$

$$C = \frac{\epsilon A}{d}$$

$$v = \frac{1}{C} \int_{t_0}^t i dt + v(t_0)$$

$$w = \frac{1}{2} C v^2$$





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PP 6.1

What is the voltage across a $3\mu\text{F}$ capacitor if the charge on one plate is 0.12 mC . How much energy is stored in the capacitor?

$$w = \frac{1}{2} C v^2 = \frac{1}{2} \times 3 \times 10^{-6} v^2$$

$$v = \frac{q}{C} = \frac{0.12 \times 10^{-3}}{3 \times 10^{-6}} = 40\text{ V}$$

$$w = \frac{1}{2} \times 3 \times 10^{-6} v^2 = \frac{1}{2} \times 3 \times 10^{-6} \times 40^2 = 2.4\text{ mJ}$$

$$q(t) = C v(t); \quad i(t) = C \frac{d}{dt} v(t)$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$



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PP 6.2

The voltage source connected across a $10\mu\text{F}$ capacitor is given as:
 $v(t) = 50 \sin 2000t\text{ V}$. Calculate $i(t)$.

$$i = C \frac{dv}{dt}$$

$$= 10 \times 10^{-6} \frac{d}{dt} (50 \sin 2000t)$$

$$= 10 \times 10^{-6} \times 50 \times 2000 \times \cos 2000t$$

$$i = \cos 2000t\text{ A}$$

$$q(t) = C v(t); \quad i(t) = C \frac{d}{dt} v(t)$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$



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PP 6.3

If the current through a $100\ \mu\text{F}$ capacitor is given as: $i(t) = 50 \sin(120\pi t)$ mA, calculate $v(1\text{ms})$ and $v(5\text{ms})$. Assume $v(0) = 0\text{ V}$

$$q(t) = Cv(t); \quad i(t) = C \frac{d}{dt} v(t)$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$w = \frac{1}{2} Cv^2 = \frac{q^2}{2C}$$

$$v = \frac{1}{C} \int_{t_0}^t i dt + v(t_0)$$

$$v(1\text{m}) = \frac{1}{C} \int_0^{1\text{ms}} 50 \sin(120\pi t) \times 10^{-3} dt + 0$$

$$= \frac{50 \times 10^{-3}}{C} \times \frac{(-\cos(120\pi t))}{120\pi} \Big|_0^{1\text{m}}$$

$$= \frac{-50 \times 10^{-3}}{100 \times 10^{-6} \times 120\pi} [\cos(120\pi \times 10^{-3}) - \cos(0)]$$

$$= -1.33 [\cos(120\pi \times 10^{-3}) - 1]$$

Change to Radians

$$v(1\text{m}) = -1.33 \times -0.07022$$

$$v(1\text{m}) = \mathbf{93.137\text{ mV}}$$

$$v(5\text{m}) = \frac{1}{C} \int_0^{5\text{ms}} 50 \sin(120\pi t) \times 10^{-3} dt + 0$$

$$= -1.33 \times \frac{(-\cos(120\pi t))}{120\pi} \Big|_0^{5\text{m}}$$

Change to Radians

$$= -1.33 [\cos(600\pi \times 10^{-3}) - 0.9999]$$

$$v(5\text{m}) = \mathbf{1.74\text{ V}}$$

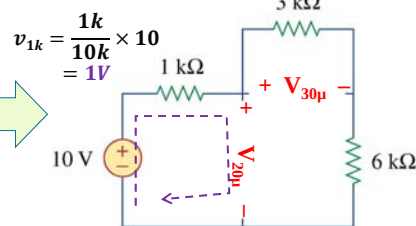
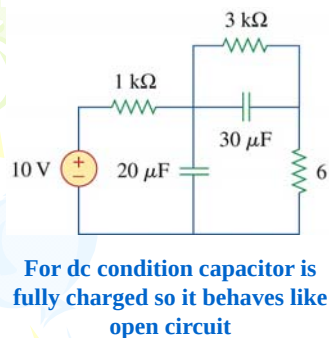


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PP 6.5 For DC conditions, calculate the energy stored in the two capacitors.



$$V_{30\mu} \text{ using voltage divider}$$

$$= \frac{3k}{1k+3k+6k=10k} \times 10$$

$$= \mathbf{3V}$$

$$w_{20\mu} = \frac{1}{2} CV_{20\mu}^2$$

$$w_{30\mu} = \frac{1}{2} CV_{30\mu}^2$$

KVL:

$$-10 + V_{1k} + V_{20\mu} = 0$$

$$\Rightarrow V_{20\mu} = \mathbf{9V}$$

$$= \frac{1}{2} 20\mu \times 9^2$$

$$= \mathbf{810\ \mu J}$$

$$w_{30\mu} = \frac{1}{2} 30\mu \times 3^2$$

$$= \mathbf{135\ \mu J}$$

$$q(t) = Cv(t); \quad i(t) = C \frac{d}{dt} v(t)$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$w = \frac{1}{2} Cv^2 = \frac{q^2}{2C}$$

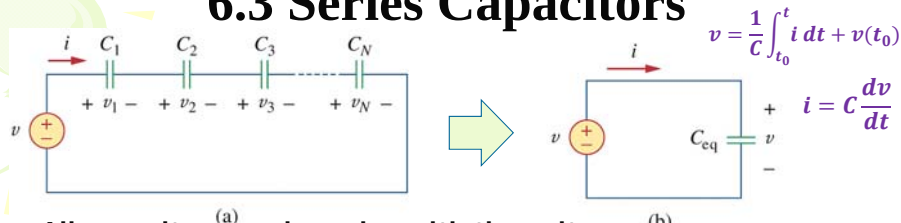


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6.3 Series Capacitors



All capacitors are in series with the voltage source

KVL: $v_1 + v_2 + v_3 + \dots + v_N - v = 0$

$$v = \frac{1}{C_1} \int_{t_0}^t i \, dt + \frac{1}{C_2} \int_{t_0}^t i \, dt + \frac{1}{C_3} \int_{t_0}^t i \, dt + \dots + \frac{1}{C_N} \int_{t_0}^t i \, dt + v(t_0)$$

$$= \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N} \right) \int_{t_0}^t i \, dt + v(t_0)$$

$$v = \left(\frac{1}{C_{eq}} \right) \int_{t_0}^t i \, dt + v(t_0)$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$$

**Series Capacitor
think Resistors in Parallel**

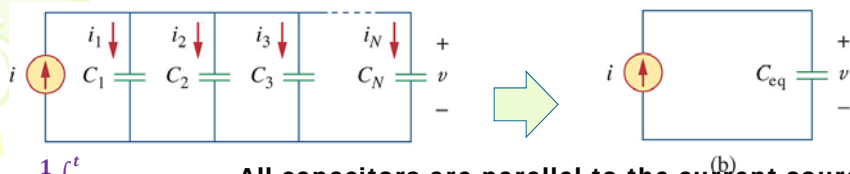


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6.3 Parallel Capacitors



All capacitors are parallel to the current source

KCL: $i_1 + i_2 + i_3 + \dots + i_N - i = 0$

$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} + \dots + C_N \frac{dv}{dt}$$

$$= (C_1 + C_2 + C_3 + \dots + C_N) \frac{dv}{dt}$$

$$i = C_{eq} \frac{dv}{dt}$$

$$C_{eq} = C_1 + C_2 + C_3 + \dots + C_N$$

**Parallel Capacitor
think Resistors in Series**



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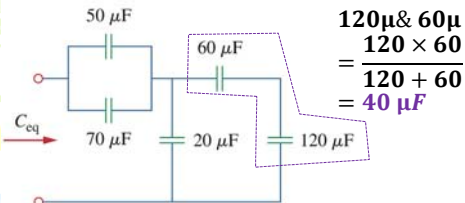
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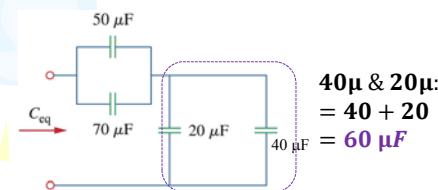
PP 6.6

Determine the equivalent capacitance

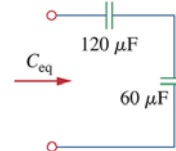
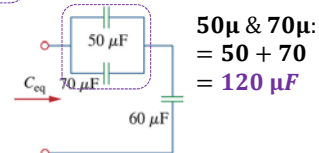
Parallel: $C_{eq} = C_1 + C_2 + C_3 + \dots + C_N$
 Series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$



$$\therefore C_{eq} = 40 \mu F$$



$$\frac{120 \times 60}{120 + 60} = 40 \mu F$$



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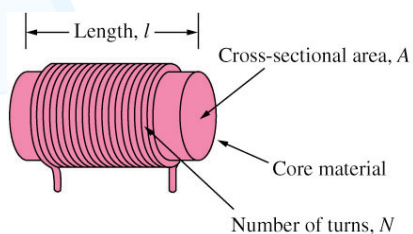
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Ch. 6 L#2

6.4 Inductors

Physical characteristics

- An inductor is a passive element designed to store energy in its **magnetic field**.



$$L = \frac{N^2 \mu A}{l} [H]$$

- An inductor consists of a coil of conducting wire.
- Core from iron, steel, plastic or air.
- μ = permeability = $\mu_r \mu_0$



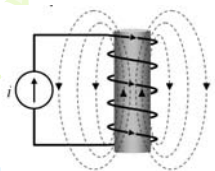
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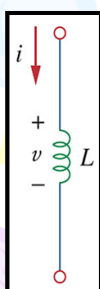
6.4 Inductors

Electrical characteristics



$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$v = L \frac{di}{dt}$$



$$p = vi = L \frac{di}{dt} i$$

$$w = \frac{1}{2} Li^2$$

Important characteristics of inductors:

- It is a **short circuit** to DC ($di/dt = 0$).
- Its **current** cannot change abruptly ($di/dt = \infty$)
- Does not dissipate energy, but stores it when charging or provides it when discharging.



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PP 6.8 The current through a 1 mH inductor is $i(t) = 20 \cos 100t$ mA. Calculate the voltage and the stored energy.

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$v = L \frac{di}{dt} \quad w = \frac{1}{2} Li^2$$

$$w = \frac{1}{2} Li^2$$

$$= \frac{1}{2} 1m (20 \cos 100t)^2$$

$$w = 0.2 \cos^2 100t \mu J$$

$$v = L \frac{di}{dt}$$

$$= 1 \times 10^{-3} \times \frac{d}{dt} \{20 \cos 100t \times 10^{-3}\}$$

$$= 20 \times 10^{-6} \times \frac{d}{dt} \{\cos 100t\} \frac{d}{dt} 100t$$

$$= 20 \times 10^{-6} \times -\sin 100t \times 100$$

$$v = -2 \sin 100t \text{ mV}$$



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PP 6.9

The voltage across a 2 H inductor is $v(t) = 10(1 - t)$

V. Determine the current through the inductor at

$t=4$ s if $i(0) = 2$ A.

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$v = L \frac{di}{dt} \quad w = \frac{1}{2} L i^2$$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$i(4) = \frac{1}{2} \int_0^4 10(1 - t) dt + 2$$

$$= 5 \left\{ \int_0^4 1 dt + \int_0^4 t dt \right\} + 2$$

$$= 5 \left\{ [t]_0^4 + \left[\frac{t^2}{2} \right]_0^4 \right\} + 2$$

$$= 5 \left[t - \frac{t^2}{2} \right]_0^4 + 2$$

$$i(4) = -18 \text{ A}$$

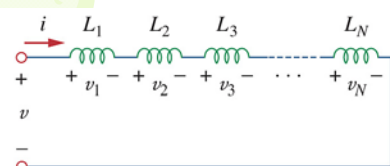


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6.4 Series Inductors



KVL:

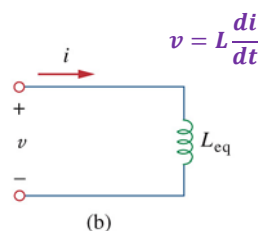
$$v = v_1 + v_2 + v_3 + \dots + v_N$$

$$v = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + L_3 \frac{di}{dt} + \dots + L_N \frac{di}{dt}$$

$$= (L_1 + L_2 + L_3 + \dots + L_N) \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2 + L_3 + \dots + L_N$$

$$v = L_{eq} \frac{di}{dt} \quad \text{— same as resistors in series}$$

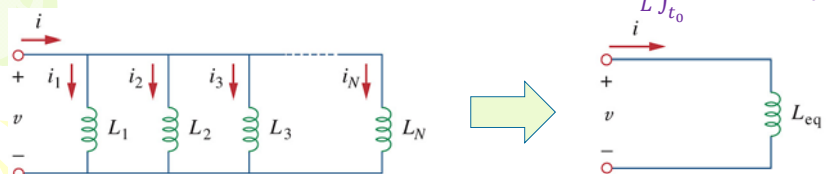


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6.4 Parallel Inductors



KCL:

$$i = i_1 + i_2 + i_3 + \dots + i_N$$

$$i = \frac{1}{L_1} \int_{t_0}^t v(t) dt + \frac{1}{L_2} \int_{t_0}^t v(t) dt + \frac{1}{L_3} \int_{t_0}^t v(t) dt + \dots + \frac{1}{L_N} \int_{t_0}^t v(t) dt + i(t_0)$$

$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N} \right) \int_{t_0}^t v(t) dt + i(t_0)$$

$$\frac{1}{L_{eq}} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N} \right)$$

$$i = \frac{1}{L_{eq}} \int_{t_0}^t v(t) dt + i(t_0)$$



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Current and voltage relationship for R, L, C

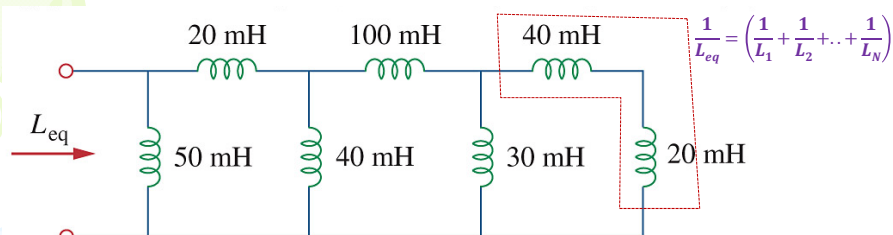
Circuit element	Resistance	Inductance	Capacitance
Units	Ohms (Ω)	Henries (H)	Farads (F)
Voltage	$v = iR$	$v = L \frac{di(t)}{dt}$	$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$
Current	$i = \frac{v}{R}$	$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$	$i(t) = C \frac{dv(t)}{dt}$
Power	$p = vi = i^2 R$	$p = L i \left(\frac{di}{dt} \right)$	$p = C v \left(\frac{dv}{dt} \right)$
Energy	$w = pt$	$w = \frac{1}{2} L i^2$	$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$
Series	$R_{eq} = R_1 + R_2 + \dots + R_N$	$L_{eq} = L_1 + L_2 + \dots + L_N$	$\frac{1}{C_{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N} \right)$
Parallel	$\frac{1}{C_{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N} \right)$	$\frac{1}{L_{eq}} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N} \right)$	$C_{eq} = C_1 + C_2 + \dots + C_N$



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PP 6.11: Calculate L_{eq} 

$$L_{eq} = (40 + 20) \parallel (30 + 100) \parallel 40 \parallel 20 \parallel 50 = 25 \text{ mH}$$

1. $40 + 20 = 60$
2. $60 \parallel 30 = 20$
3. $20 + 100 = 120$
4. $120 \parallel 40 = 30$
5. $30 + 20 = 50$
6. $50 \parallel 50 = 25 \text{ mH} = L_{eq}$



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PP 6.12

If $i_1(t) = 0.6e^{-2t}$ A and $i(0) = 1.4$ A, find

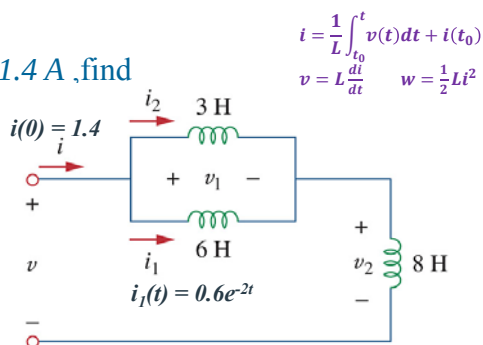
- a) $i_2(0)$
- b) $i_2(t)$ and $i(t)$
- c) $v_1(t)$, $v_2(t)$ and $v(t)$

$$i_2(t) = \frac{1}{L_{3H}} \int v_1 dt + i_2(0)$$

$$i_2(t) = \frac{1}{3} \int_0^t v_1 dt + 0.8$$

$$\begin{aligned} v_1(t) &= L_{6H} \frac{di_1}{dt} \\ &= 6 \frac{d}{dt} \{0.6e^{-2t}\} \\ &= 6 \times -2 \times 0.6e^{-2t} \\ &= -7.2e^{-2t} \text{ V} \end{aligned}$$

$$\begin{aligned} i_2(t) &= \frac{1}{3} \int_0^t -7.2e^{-2t} dt + 0.8 = \frac{-7.2}{3} \left[\frac{1}{-2} e^{-2t} \right]_0^t + 0.8 = 1.2[e^{-2t} - 1] + 0.8 \\ &= 1.2e^{-2t} - 0.4 \text{ A} \end{aligned}$$



$$\begin{aligned} \text{KCL: } i(0) &= i_1(0) + i_2(0) \\ 1.4 &= 0.6e^{-2(t=0)} + i_2(0) \\ i_2(0) &= 1.4 - 0.6 \\ &= 0.8 \text{ A} \end{aligned}$$



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PP 6.12 continued ...

If $i_1(t) = 0.6e^{-2t}$ A and $i(0) = 1.4$ A, find

(b) $i_2(t)$ and $i(t)$

(c) $v_1(t)$, $v_2(t)$ and $v(t)$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$v = L \frac{di}{dt} \quad w = \frac{1}{2} L i^2$$

$$i_2(0) = 0.8 \text{ A}$$

$$v_1(t) = -7.2e^{-2t} \text{ V}$$

$$i_2(t) = 1.2e^{-2t} - 0.4 \text{ A}$$

KCL:

$$i(t) = i_1(t) + i_2(t)$$

$$= 0.6e^{-2t} + 1.2e^{-2t} - 0.4 \text{ A}$$

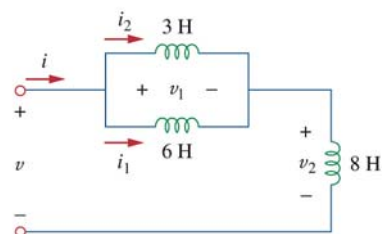
$$= 1.8e^{-2t} - 0.4 \text{ A}$$

$$v_2(t) = L_{8H} \frac{di}{dt}$$

$$= 8 \frac{d}{dt} \{1.8e^{-2t} - 0.4\}$$

$$= 8 \times -2 \times 1.8e^{-2t}$$

$$= -28.8e^{-2t} \text{ V}$$



KVL:

$$v(t) = v_1(t) + v_2(t)$$

$$= -7.2e^{-2t} - 28.8e^{-2t}$$

$$= -36e^{-2t} \text{ V}$$



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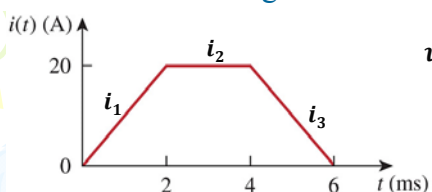
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Problem 6.40

The current through a 10 mH inductor is shown below.

Determine the voltage across the inductor at $t = 1, 3$ and 5 ms.



$$v(t) = L \frac{di}{dt} = 10\text{m} \frac{d}{dt} i(t)$$

$$i(t) = \frac{\Delta i}{\Delta t} t + c$$

$$i_1(t) = \frac{20 - 0}{0.002 - 0} t + 0 = 10t \text{ kA}$$

$$i(t) = \begin{cases} i_1 = 10t \text{ kA} & 0 < t < 2\text{m} \\ i_2 = 20 \text{ A} & 2\text{m} < t < 4\text{m} \\ i_3 = -10t \text{ kA} & 4\text{m} < t < 6\text{m} \end{cases}$$

$$v(1\text{m}) = 0.01 \frac{di_1}{dt}$$

$$= 0.01 \frac{d}{dt} 10t \text{ (kV)}$$

$$= 100\text{V}$$

Or

$$= 0.01 \frac{\Delta i_1}{\Delta t}$$

$$= 0.01 \frac{20 - 0}{2\text{m} - 0} = 100\text{V}$$

$$v(3\text{ms}) = 0.01 \frac{di_2}{dt}$$

$$= 0.01 \frac{d}{dt} (0) = 0 \text{ V}$$

$$v(5\text{ms}) = 0.01 \frac{di_3}{dt}$$

$$= 0.01 \frac{d}{dt} (-10t) \text{ (kV)} = -100\text{V}$$

Or

$$= 0.01 \frac{\Delta i_3}{\Delta t}$$

$$= 0.01 \frac{0 - 20}{6\text{m} - 4\text{m}} = -100\text{V}$$



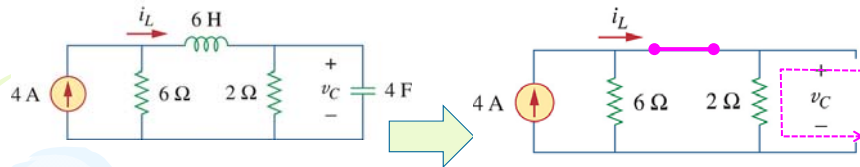
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PP 6.10

Determine v_C , i_L and the energy stored in both the capacitor and the inductor for the DC circuit below.



For dc condition:

➤ Capacitor is fully charged it behaves like open circuit.

➤ Inductor behaves as short-circuit

$$i_L = i_{2\Omega} = 4 \frac{6}{6 + 2} = 3 \text{ A} \quad v_C = v_{2\Omega} = 2i_L = 6 \text{ V}$$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

$$v = L \frac{di}{dt} \quad w = \frac{1}{2} Li^2$$

$$w = \frac{1}{2} Li^2 = 0.5 \times 6 \times 3^2 = 27 \text{ J}$$

$$w = \frac{1}{2} Cv^2 = 0.5 \times 4 \times 6^2 = 72 \text{ J}$$

$$q(t) = Cv(t); \quad i(t) = C \frac{dv}{dt}$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

$$w = \frac{1}{2} Cv^2 = \frac{q^2}{2C}$$



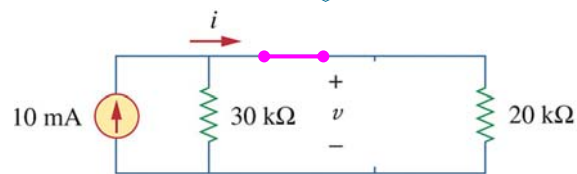
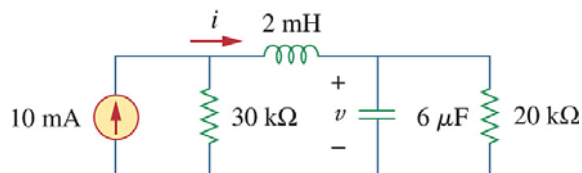
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Problem 6.48

Under steady state dc conditions, find i and v .



$$i = i_L = 10 \text{ mA} \times \frac{30}{30 + 20} = 6 \text{ mA}$$

$$v = v_C = 6 \text{ mA} \times 20 \text{ k}\Omega = 120 \text{ V}$$



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